

Supplementary Information for

Recipe determinants of quantization fragility under image corruption:

paired evidence and a regression-head localization in object detectors

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S1 Purpose and scope

This supplement documents the frozen protocol details, estimands, corruption parameters, calibration schedules, and artifact contract supporting the main article. All reported values are generated deterministically from retained ledgers. Section S8 additionally reports per-family ΔE diagnostics that support—but do not extend—the main-text claims.

S2 Estimands

For dataset d , model m , corruption condition c , and a pair of treatments (A, B) sharing the same evaluator:

$$G_0 = \text{AP}_A(\text{J95}) - \text{AP}_B(\text{J95}), \quad (\text{S1})$$

$$G_{\text{corr}} = \overline{\text{AP}_A(c) - \text{AP}_B(c)} \quad (\text{equal weight over the 12 cells}), \quad (\text{S2})$$

$$\Delta E = G_{\text{corr}} - G_0. \quad (\text{S3})$$

J95 denotes the codec-matched clean control: the J95 clean control and every corrupted input are terminally encoded as JPEG quality 95 with chroma subsampling disabled, so the clean control and the corrupted inputs differ only in the named corruption transform, not in container encoding; the separate original-clean cell keeps the dataset-native bytes; it checks control fidelity and anchors the clean-accuracy comparisons behind the RetinaNet collapse and recovery claims, while the paired interaction uses the J95 basis. $\Delta E > 0$ means the A-minus-B gap widens under corruption; $\Delta E < 0$ means it narrows. A narrowing gap near a shared accuracy floor is not read as robustness: absolute corrupted AP is reported alongside every interaction (main text, §4.1–§4.3).

S3 Frozen condition schedule

Each dataset×model block evaluates 14 conditions on identical image ordering for quantized arms (13 for the RetinaNet FP32 and INT8-legacy arms that predate codec control):

- clean (original bytes) and J95 (JPEG-95 clean control);
- Gaussian noise: $\sigma \in \{8, 20, 40\}$;
- motion blur: kernel lengths $\{5, 13, 23\}$;
- synthetic fog: (strength, radius) $\in \{(0.18, 6), (0.38, 12), (0.62, 20)\}$;
- JPEG compression: qualities $\{70, 40, 15\}$.

Corruption parameters and the JPEG-95 terminal encoding are frozen in versioned configuration `corruptions.json`, whose SHA-256 is recorded as `generator_config_sha256` in the corruption-aware and codec-control calibration manifests; the frozen protocol file `nn_paired_protocol_v1_20260924` fixes the condition list and

evaluation contract on top of that generator. The same transform code produces both the evaluation caches and the corruption-aware calibration images.

S4 Treatments and arm registry

Table S1: RetinaNet arm registry. “Calibration” lists the input distribution seen by entropy calibration; “sites” lists the quantized region.

Arm	Precision	Quantized sites	Calibration input
FP32	FP32	none	—
FP8-matched	FP8	full graph	clean, matched preprocessing
INT8-legacy	INT8	full graph	mismatched preprocessing (diagnostic)
INT8-matched	INT8	full graph	clean, 512 images
INT8-q95calib	INT8	full graph	J95 re-encodes, 512 images
INT8-selective	INT8	all except <code>/head/regression.head/</code>	clean, 512 images
INT8-sel-q95calib	INT8	selective	J95 re-encodes, 512 images
INT8-corruptcalib	INT8	full graph	fold-1 corruptions (noise, JPEG), 512 images
INT8-sel-corruptcalib	INT8	selective	in-family corruptions, 512 images
INT8-cc2calib	INT8	full graph	fold-2 corruptions (fog, motion blur), 512 images
INT8-sel-cc2calib	INT8	selective	fold-2 corruptions, 512 images
INT8-wonly	INT8	full graph; head <i>acts.</i> fp16	clean, 512 images
INT8-aonly	INT8	full graph; head <i>weights</i> fp16	clean, 512 images
FP8-legacy	FP8	full graph	pre-study build (context only)

The W-only and A-only arms are post-hoc QDQ-graph rewires of the frozen INT8-matched graph (Section 3.7 of the main text): under the TensorRT `int8-entropy` build policy every non-quantized operation executes in fp16, so the restored operand computes in fp16 rather than literal fp32, and the bypassed head weight initializers are themselves fp16 tensors.

Table S2: FCOS-R50-FPN replication arm registry. The selective exclusion regex `~/head/regression.head/` covers the regression tower and the `bbox_ctrness` output, so the selective arm also dequantizes centerness—because the score is $\sqrt{\sigma(\text{cls}) \cdot \sigma(\text{ctr})}$, the intervention is not treatment-identical to the RetinaNet selective arm, which touches regression outputs only. Calibration identical to the RetinaNet matched recipe.

Arm	Precision	Quantized sites	Calibration input
FP32	FP32	none	—
FP8-matched	FP8	full graph	clean, matched preprocessing
INT8-matched	INT8	full graph	clean, 512 images
INT8-selective	INT8	all except regression tower + centerness	clean, 512 images

All INT8/FP8 engines are built by TensorRT 11.1.0.106 from ONNX graphs quantized by NVIDIA ModelOpt with entropy calibration (512-image budget; the study-built registries record version 0.45.0 and the pre-study YOLO11 ONNX registries predate that field; the diagnostic sweeps vary the estimator and include arms marked “128-img” in Table 4 of the main text, which use the earlier 128-image subset) with a 4,096 MiB builder workspace. Builder policy is recorded per engine and differs by program: the YOLO11 engines were built by `trtexec` before the frozen contract and carry TensorRT 11.1 defaults (TF32 enabled); the RetinaNet quantization arms built for this study use the TensorRT Python builder with TF32 disabled, while the RetinaNet FP32 reference and the FP8/INT8-legacy arms are pre-study `trtexec` builds with TF32 enabled (the ΔE contrasts only join quantized arms built under the same policy); the four FCOS arms, FP32 included, are TensorRT Python builder builds with TF32 disabled, so the FCOS block is internally policy-matched but not builder-identical to the pre-study YOLO11 engines. Calibration and evaluation images are disjoint. Checkpoints are one per dataset $\times$ model cell, and their selection provenance is disclosed: the KITTI and VOC YOLO11n/m/x models were fine-tuned from Ultralytics pretrained weights under the pilot program (`*_train_v1` runs, dataset YAMLS: KITTI `val: images/val`, VOC `val: images/test2007`), with the best checkpoint taken on the same partition that serves as the evaluation universe; COCO YOLO11 uses the official pretrained checkpoint without a dataset fine-tune; the torchvision RetinaNet-R50-FPN-v2 model is fine-tuned per dataset for 100 epochs at 640-pixel input (learning rate 10^{-4} , weight decay 10^{-4} , deterministic seed 20260807), selecting the best checkpoint by minimum validation loss on the same evaluation partition. FCOS-R50-FPN is trained under the same profile and checkpoint rule. All four RetinaNet/FCOS selected checkpoints are early picks: run-log epochs 21 (KITTI) and 5 (VOC) for RetinaNet, 14 (KITTI) and 10 (VOC) for FCOS (epoch counts here are the run log’s 1-indexed values; the checkpoints store

the same point 0-indexed). The KITTI/VOC YOLO11 models instead select late best-fitness checkpoints (run epochs 65–100; recorded in `phase_a_arms/training/yolo11.best_fitness_epochs.json`) under Ultralytics’ own rule, and the COCO YOLO11 models are the official pretrained weights with no study-side selection. Both FCOS runs completed all 100 scheduled epochs. On VOC the FCOS training loss diverged to NaN at epoch 70 (validation loss stayed finite through epoch 92, then also diverged); the epoch-10 selection point predates the divergence by 60 epochs, and no intervening epoch beat its validation minimum, so the divergence did not affect which checkpoint was exported. The training manifest records the schedule and selection rule but not the loss history; the per-epoch training/validation loss curves are shipped in the support package (`phase_b_results/training/` for RetinaNet, `phase_h_fcos_results/training/` for FCOS), and the selected epochs match the checkpoints’ stored epoch fields (raw run logs are not redistributed; each curve record carries its source log’s SHA-256). Per-run training manifests are hash-bound; the evaluation-partition overlap is a disclosed selection caveat (see the main text’s limitations). The frozen condition schedule is 14 conditions per arm; the RetinaNet FP32 and legacy arms predate codec control and carry 13.

S5 Inference and evaluation contract

YOLO11 blocks: 640-pixel letterbox input, confidence threshold 0.001, class-wise NMS at IoU 0.7, at most 300 detections. RetinaNet blocks: 640-pixel letterbox input, confidence threshold 0.05, batched NMS at IoU 0.5, at most 300 detections. FCOS blocks: 640-pixel letterbox input, confidence threshold 0.05 on the centerness-weighted score, at most 1,000 pre-NMS candidates per feature level, batched NMS at IoU 0.5, at most 300 detections—the RetinaNet constants reused post hoc and frozen for FCOS at the RetinaNet values, and not torchvision’s FCOS defaults (score threshold 0.2, NMS IoU 0.6, 100 detections per image); the decoder is a reimplement of torchvision’s post-processing, validated detection-for-detection against torchvision’s own pipeline on 32 seeded synthetic head-output draws (`phase_h_fcos_results/decoder_validation.json`: 9,600 detections, maximum box and score deviation exactly zero). The check saturates the 300-detection cap on every draw, uses a square 640-pixel input at scale 1.0, and covers the 8-class KITTI head only—it validates the post-processing arithmetic, not letterbox un-mapping, sparse-output behavior, or end-to-end engine parity. All evaluations use the COCO protocol via `pycocotools` with `maxDets=100`; VOC and KITTI annotations were converted once to COCO format and frozen. Decoder constants are listed per family in the paragraph above—identical for RetinaNet and FCOS—and held fixed within each; cross-family comparisons are comparisons of recipe bundles, never of formats alone.

S6 Corruption-aware calibration schedule

The intervention calibration sets reuse the same 512 images as the matched clean-calibration manifests. Each image is deterministically assigned—via a SHA-256-keyed hash of its identifier—to exactly one of six in-family cells, Gaussian noise or JPEG at severities 1, 3, or 5 (approximately 77–102 images per cell). Assigned corruptions use the frozen transform parameters above, and the outputs are terminally encoded at JPEG quality 95, matching the evaluation input contract. In this first fold fog and motion blur never appear in calibration and form the held-out split. A complementary second fold (schedule `nn-corruptcalib-fold2-v1`, frozen before any of its runs) assigns the same images to fog or motion blur at severities 1, 3, or 5 (75–95 images per cell) and holds out Gaussian noise and JPEG. Its pre-specified rules are (i) a held-out change from INT8-matched whose interval excludes zero (either direction counts; the main text reads it through the directional fold-1 criterion) and (ii) a positive held-out ΔE with an interval excluding zero, against both INT8-matched and INT8-q95calib.

The codec-matched control manifests encode the same 512 images at JPEG quality 95 with no named corruption, isolating codec adaptation from corruption coverage.

S7 Paired bootstrap

Within each dataset×model block, one deterministic schedule of $B = 2,000$ resamples of image indices is shared across all arms and all conditions, so every contrast is evaluated inside common-image samples. Reported point estimates are full-sample plug-in values; reported intervals are the 2.5th–97.5th percentiles of the bootstrap distribution of each contrast. Intervals cover image resampling only; they do not cover training-seed, calibration-draw, engine-build, or corruption-realization variance, and image-level resampling treats frames as independent (KITTI frames from shared driving sequences are correlated, so those intervals may be optimistic).

Two-sided bootstrap p -values are $p = \min\{1, 2 \min[\Pr(d \leq 0), \Pr(d \geq 0)]\}$ (two-sided resolution 1/1000) and are Holm-adjusted across the nine paired YOLO11 blocks. Heterogeneity of ΔE across those blocks uses Cochran’s $Q = \sum_i w_i (\hat{\Delta}_i - \bar{\Delta}_w)^2$ with $w_i = 1/\hat{\text{se}}_i^2$ from the bootstrap draws, $I^2 = \max\{0, (Q - \text{df})/Q\}$, and the DerSimonian–Laird between-block standard deviation τ . Blocks within a dataset share images, so their estimates may be correlated; the cross-block covariance is not estimated, and treating the blocks as independent is an approximation whose error direction is not established for these differences-of-differences (positive covariance would make Q

and the pairwise z statistics conservative, but this is an assumption, not a verified property). RetinaNet and intervention intervals are per contrast.

Point estimates are full-sample plug-in values computed from the metric records of each cell (for COCO, from the frozen 2,000-image subset evaluation); bootstrap draws supply intervals, standard errors, and p -values only. For the intervention, the corruption-specific component on the held-out family is $\Delta E_{\text{held-out}} = (\text{AP}_A - \text{AP}_B)_{\text{held-out}} - (\text{AP}_A - \text{AP}_B)_{\text{J95}}$, the same paired construction as the main estimand. This decomposition is interpretive: it assumes an additive, parallel-shift model in which a calibration-induced level shift on clean inputs transfers comparably to corrupted inputs. Under an interaction model (where the intervention also changes slope across conditions), $\Delta E_{\text{held-out}}$ mixes that interaction with any clean-level effect; it is not a causal identification theorem. We therefore report the J95-clean and held-out levels alongside it rather than relying on the composite alone.

S8 Sensitivity analyses

Table S3 resolves the paired ΔE of each YOLO11 block per corruption family (each entry is the family-mean FP8–INT8 gap minus the J95 gap) and gives the leave-one-out exclusion of the `motion_blur-s5` severity wall. The sign heterogeneity is also *within* blocks: on VOC-x the JPEG family contributes -1.07 AP while Gaussian noise and motion blur contribute $+0.69$ and $+0.67$ AP; on COCO-m JPEG ($+1.43$) and motion blur ($+0.99$) contribute positively while noise contributes a null negative (-0.36). Motion blur is the most block-variable family of the four (across-block standard deviation ≈ 1.1 AP population, ≈ 1.2 sample), with JPEG close behind (≈ 1.0); neither is uniformly harmful or helpful. Excluding the severity wall leaves the cross-block heterogeneity essentially unchanged ($Q \approx 60.2$, $I^2 \approx 87\%$) and every Holm-surviving block’s sign intact; the two KITTI blocks that do not survive Holm drift (YOLO11n’s null point estimate flips from -0.01 to $+0.15$, and YOLO11m’s unadjusted-significant negative turns null). The interaction pattern is therefore not an artifact of the saturated cell, although the marginal KITTI estimates are sensitive to it.

The same leave-one-out sensitivity applied to the intervention contrasts changes no frozen-criterion verdict; two secondary codec-control contrasts do move across the zero boundary on VOC (both on the held-out basis): INT8-corruptcalib minus INT8-q95calib goes from $+0.13$ $[-0.02, +0.28]$ to $+0.21$ $[+0.04, +0.38]$, and INT8-sel-q95calib minus INT8-selective goes from -0.11 $[-0.22, +0.01]$ to -0.15 $[-0.29, -0.01]$. Both are read with the main text’s no-multiplicity caveat.

The frozen configuration files are hash-identified: `nn_corruptcalib_v1_20260924.json` SHA-256 `f51011ca5d666b89...` and `nn_corruptcalib_fold2_v1_20260925.json` SHA-256 `5f2d122ae0a52d5f...` (full hashes in the archive manifest).

Freeze timing is established by local-file evidence rather than an external registry: the versioned configuration files’ modification times and the `created_at_utc` fields in the calibration and quantized-ONNX manifests place the configurations before their arms’ first inference (run manifests carry no timestamp field, and the git history records joint archival of configs and outcomes, not pre-inference ordering). The fold-1 file was written 2026-09-25 (its name retains the design date 20260924, and its `title` field says “frozen 2026-09-24”—the title records the design-freeze day while the serialized file postdates it by one day); the q95 codec-control schedules were added after the fold-1 results were computed—the attempt identifier `nn_q95calib_v1_20260924` keeps the design-date convention and postdates the fold-1 runs in manifest time. The fold-2 primary rule is quoted verbatim: “`int8-cc2calib512 minus int8-matched512 on the fold-2 held-out mean (gaussian_noise+jpeg), paired bootstrap B=2000, same seed namespace as Phase B: CI95 excluding 0 => fold-1 frozen criterion replicated on fold 2`”; we read “replicated” through the directional fold-1 criterion it cites, so a significantly negative change counts as failure.

Table S3: Per-family ΔE (AP points; family-mean FP8–INT8 gap minus J95 gap) and the leave-one-out ΔE excluding `motion_blur-s5`, per dataset×model block. * interval excludes zero.

Dataset	Model	Fog	Gaussian noise	JPEG	Motion blur	ΔE excl. mb-s5
KITTI	YOLO11n	$+1.11^*$	$+0.41$	$+0.17$	-1.71^*	$+0.15$ $[-0.51, +1.03]$
KITTI	YOLO11m	$+0.38$	-0.44	$+0.02$	-2.68^*	-0.35 $[-0.98, +0.29]$
KITTI	YOLO11x	-0.40	-1.98^*	-0.43	$+0.03$	-0.65^* $[-1.08, -0.29]$
VOC	YOLO11n	$+0.35^*$	$+0.28$	$+0.27$	$+0.34^*$	$+0.34^*$ $[+0.09, +0.62]$
VOC	YOLO11m	-0.03	-0.48^*	-1.23^*	-0.56^*	-0.53^* $[-0.76, -0.31]$
VOC	YOLO11x	$+0.26^*$	$+0.69^*$	-1.07^*	$+0.67^*$	$+0.13$ $[-0.06, +0.35]$
COCO-2k	YOLO11n	$+0.17$	-0.22	-0.56^*	-0.69^*	-0.28 $[-0.63, +0.05]$
COCO-2k	YOLO11m	$+0.36^*$	-0.36	$+1.43^*$	$+0.99^*$	$+0.60^*$ $[+0.27, +0.97]$
COCO-2k	YOLO11x	$+0.15$	-0.28	-2.10^*	$+0.13$	-0.59^* $[-1.00, -0.26]$

S9 Evidence layers

Diagnostic arms (the RetinaNet decomposition: head-only and all-except-region variants) are single-point recipe sweeps produced during development and carry no intervals. Frozen paired evidence is the 14-condition paired protocol; its freeze covers the estimands, condition schedule, and bootstrap schedule (the nine-block Holm adjustment lives in the analysis code, not the frozen configuration), while the YOLO11 engines and evaluation partitions predate it and are shared with the historical layer. Intervention evidence is the corrupt-calibration experiment together with its codec-matched controls: the two corruption-calibration folds were frozen prospectively before their engines were built, while the fold-1 codec-control arms are post-hoc. Historical exploratory results (a 144-cell grid over four datasets including TT100K, computed on the same engines and estimand under a wider scope and a different analysis plan) are reported only as context and are never pooled with the frozen paired estimates. Post-hoc extensions are the operand-level factorial inside the RetinaNet regression head (W-only and A-only rewires of the frozen matched graph) and the FCOS-R50-FPN replication: both were designed and executed after the frozen paired results were known, are evaluated on the same condition schedule with the same paired bootstrap estimand, and are reported as post-hoc extensions rather than frozen evidence. The FCOS phase package ships its own provenance (per-engine build manifests, quantized-ONNX manifests, per-run training manifests, and the decoder-validation record above) under `phase_h_fcos_results/`.

S10 Artifact and audit boundary

Manuscript tables and figures are generated deterministically from frozen ledgers by `analysis/build_nn_tables.py`. Engine registries, calibration manifests (including the corrupt-calibration cell-assignment schedule), input manifests, prediction manifests, metric records, and paired-bootstrap draw caches are SHA-256 hash-bound and retained in the accompanying compact-evidence package, deposited in the versioned research archive under the concept DOI <https://doi.org/10.5281/zenodo.22031663> (currently resolving to the prior CVIU-era release lineage; the Neural Networks release mints on publication). Raw benchmark datasets, trained checkpoints, built TensorRT engines, and raw predictions are not redistributed; their provenance and regeneration path is documented in the archive. One binding asymmetry is disclosed for transparency: corrupted, codec-control, and YOLO11 original-clean cells bind each run to a content-addressed input manifest covering the encoded bytes (the YOLO11 clean runs bind the per-image clean manifests such as `kitti_val_clean_ultralytics_v1`), whereas the RetinaNet original-clean cells bind the frozen dataset split through its annotation SHA-256 (`clean-root:<hash>`). The matching per-image clean manifests exist in the archive and establish the reference bytes of each frozen split, but they are not bound into the RetinaNet clean run records; those runs therefore do not carry an execution-time byte-level binding for the clean inputs, only the annotation-level identifier.

S11 Minimum reporting checklist for paired corruption studies

- record the complete treatment recipe (checkpoint, quantizer coverage, calibration list and preprocessing, builder settings, decoder), not just the datatype label;
- pair treatments on identical image bytes, including terminal encoding;
- subtract the matched clean gap before reading a corruption interaction;
- report absolute corrupted AP and the clean gap alongside every interaction, so floor effects cannot masquerade as robustness;
- keep calibration corruption families disjoint from held-out evaluation families when claiming calibration-based robustness;
- carry codec-matched calibration controls before attributing calibration-intervention gains to corruption coverage;
- flag severity-wall cells where every treatment including FP32 is at the floor, and report a leave-one-out sensitivity.